

GabAIRa Care: An AI-Driven Medication Management Application with IoT Pillbox Integration for Improved Adherence

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Abstract—Medication non-adherence remains a critical challenge in chronic disease management, particularly among patients with hypertension, diabetes, and cardiovascular conditions. This paper presents GabAIRa Care: *Gabay na may Pag-alaga, Kahit Saan, Kahit Kailan*, an XGBoost-assisted cross-platform medication adherence management system integrated with IoT-enabled manual dispensing and caregiver synchronization features. The system employs a multi-output XGBoost classifier trained on 10,000 synthetic patient records to simultaneously predict adherence risk (Low/High) and recommended reminder action (Send Now / Delay / Snooze), achieving 100% accuracy on the Risk Model and 100% weighted-average accuracy on the Action Model with macro-average F1-scores of 1.00 and 0.98, respectively. The proposed solution is accessible on Android platform via React Native and Expo, and communicates with an ESP32-based smart pill dispenser through REST API synchronization backed by a Neon PostgreSQL database. A Linear Algorithm performs real-time schedule scanning while Dynamic Programming progressively optimizes future reminder timing based on historical adherence behavior. User Acceptance Testing conducted with four respondents, three students and one professional pharmacist, yielded a 100% task completion rate for all Patient Role test cases and a 99.1% pass rate for Caretaker Role test cases. ISO/IEC 25010 evaluation produced Highly Acceptable ratings in Usability (4.80), Functionality Suitability (4.56), and Reliability (4.50), confirming that GabAIRa Care effectively addresses medication management challenges for both patients and caregivers.

Index Terms—medication adherence, XGBoost, IoT, pill dispenser, caregiver synchronization, REST API, dynamic programming, chronic disease management

I. INTRODUCTION

A. Background of the Study

Medication adherence, the extent to which patients follow prescribed treatment schedules, is one of the most persistent problems in modern healthcare. The World Health Organization reports that only approximately 50% of patients with long-term therapies consistently take their medications as prescribed, directly contributing to disease progression, increased hospitalizations, and preventable deaths [1]. In the Philippines, local studies in Davao City confirm this trend, finding significant non-adherence among hypertensive and diabetic patients due to forgetfulness, complex multi-drug regimens, and inadequate caregiver support [2].

Many caregivers still depend on handwritten notes, basic phone alarms, or ordinary pill organizers to track medication schedules. While these methods may suffice for simple routines,

they frequently result in missed doses or incorrect medication intake, especially when managing multiple prescriptions simultaneously. Existing digital reminder tools are largely static: they notify patients at fixed times but cannot adapt to behavioral patterns, synchronize with caregivers in real time, or physically dispense medication [6]. Fully automated pharmaceutical dispensers exist but are often classified as medical devices, carry high costs, and remain inaccessible to ordinary households.

This gap motivates the development of GabAIRa Care, a system that integrates Artificial Intelligence (AI), Internet of Things (IoT), and cross-platform mobile technologies into a unified medication adherence solution. XGBoost has demonstrated strong performance in structured healthcare prediction tasks, making it suitable for analyzing patient adherence patterns and generating adherence risk insights to guide reminder delivery [5]. A REST API architecture provides a reliable, stateless communication layer that synchronizes the ESP32-based pillbox, the mobile clients, and the cloud database without requiring a persistent broker connection.

B. Problem Statement

This study aims to design, develop, and evaluate GabAIRa Care, an AI-driven, IoT-integrated medication adherence management system for patients and caregivers. Specifically, the study seeks to:

- (1) classify adherence risk and determine the optimal reminder action using a trained XGBoost model;
- (2) automate physical medication dispensing through an ESP32-based IoT smart pill box synchronized with the mobile application via REST API;
- (3) enable real-time caregiver monitoring and bidirectional notifications between patients and caregivers; and
- (4) implement a Linear Algorithm for real-time schedule scanning and a Dynamic Programming module for adaptive reminder optimization and drug interaction checking based on historical behavior.

C. Significance of the Study

GabAIRa Care delivers direct and measurable benefits across multiple stakeholder groups. For patients managing chronic illness, the system provides timely, AI-optimized medication reminders that reduce missed doses and improve treatment outcomes. For family caregivers and professional caretakers, realtime monitoring and dashboards reduce the burden of manual tracking and provide early warning of missed or delayed doses. For healthcare providers, adherence logs and risk classifications support evidence-based patient management without requiring dedicated clinical infrastructure. For the academic and research

community, the study demonstrates a reproducible architecture combining AI-driven behavioral classification, IoT hardware synchronization, and cross-platform development for medication adherence support in a Philippine healthcare context.

D. Scope and Limitations

The scope of GabAIRa Care covers: (1) Android-based platform support using React Native and Expo; (2) medication scheduling, push notification reminders, and caregiver synchronization; (3) a multi-output XGBoost classifier trained on a 10,000-record synthetic dataset for adherence risk and reminder-action prediction; (4) an ESP32-based IoT smart pillbox with buzzer and LED alerts synchronized via REST API; and (5) a Linear Algorithm for real-time schedule scanning and a Dynamic Programming module for drug interaction checking and reminder optimization.

The study is limited in that the XGBoost model was trained exclusively on synthetic data rather than longitudinal real-patient records, and the IoT pill dispenser is an academic prototype rather than a certified medical device. The system does not replace professional medical consultation or healthcare supervision, and it requires stable internet connectivity for REST API communication and IoT synchronization.

E. Algorithm Concepts

GabAIRa Care integrates three main algorithmic components across its software and AI layers.

The first is the **multi-output XGBoost classifier**, implemented as a `MultiOutputClassifier` wrapping an `XGBClassifier` from the XGBoost library. The model is trained on 12 behavioral and demographic features to simultaneously predict two target labels: `label risk` (binary: Low/High) and `label action` (multiclass: Send Now / Delay / Snooze). Hyperparameters were configured as `n_estimators=100`, `max_depth=4`, `learning_rate=0.1`, and `eval_metric='mlogloss'`, selected for their balance between predictive accuracy and inference latency suitable for real-time mobile deployment.

The second is the **Linear Algorithm** for real-time schedule scanning. At each reminder check interval, the algorithm sequentially scans the patient's complete medication list to identify which doses are currently due, operating in linear time relative to the number of active medications and guaranteeing that no scheduled dose is overlooked.

The third is the **Dynamic Programming** module, applied to two sub-problems. For drug interaction checking, the algorithm builds on previously solved interaction sub-checks to evaluate combined medication regimens efficiently rather than recomputing each pairwise check from scratch. For medication name reconciliation, it applies edit-distance-based fuzzy matching to correct minor spelling variations entered by patients or caregivers. The same historical-adherence data structure is also used to progressively shift future reminder times toward hours of demonstrated higher compliance for a given patient.

II. RELATED WORKS

The following review examines literature relevant to GabAIRa Care's AI-driven risk classification, IoT pillbox synchronization, and medication scheduling algorithms. Table I synthesizes each study's technical approach, key findings, and the limitation that

GabAIRa Care addresses.

A. Medication Adherence Systems

Medication non-adherence is a globally documented healthcare concern. The World Health Organization's foundational review established that only around half of patients on long-term therapies adhere consistently, a pattern echoed in the Davao City context, where hypertensive and diabetic patients reported significant non-adherence linked to forgetfulness and inadequate caregiver support [1], [2]. Mobile-based adherence systems have been shown to improve medication consistency among patients with chronic illnesses, particularly when integrated with real-time reminders and caregiver support, though pilot deployments have generally remained notification-only without physical dispensing or confirmation mechanisms [4].

B. Artificial Intelligence in Healthcare

AI-assisted healthcare systems are increasingly capable of analyzing patient behavior, detecting missed-dose patterns, and generating adherence risk insights from historical data [7]. XG-Boost, the gradient-boosted decision tree framework underlying GabAIRa Care's classification layer, has demonstrated strong performance and computational efficiency on structured tabular data across domains, making it well-suited for adherence risk and reminder-action prediction at mobile-deployable latency [5].

C. IoT Healthcare Devices and REST API Communication

IoT-enabled medication systems utilizing microcontrollers, sensors, LEDs, buzzers, and automated dispensing mechanisms have been shown to reduce missed doses and improve adherence, particularly among elderly patients requiring continuous monitoring. Digital solutions for informal caregivers remain fragmented across separate apps and devices, with few unifying physical dispensing, mobile notification, and caregiver dashboards in a single platform [6]. GabAIRa Care's REST API architecture addresses this by providing one stateless, request-response communication layer between the ESP32 pillbox, the mobile clients, and the Neon PostgreSQL database.

D. Mobile Health Technologies in the Philippine Context

Mixed-methods research on mobile adherence support in the Philippines found that interactive mobile reminders can meaningfully improve patient engagement when adapted to local healthcare access patterns, though existing studies stop short of integrating physical IoT dispensing or AI-driven behavioral risk scoring [8].

E. Scheduling Algorithms in Healthcare Systems

Intelligent medication schedule generation systems demonstrate that algorithmic approaches can reliably produce universal reminder schedules from prescription data [9]. GabAIRa Care extends this line of work by pairing a Linear Algorithm for real-time due-dose scanning with a Dynamic Programming approach, grounded in standard algorithmic complexity theory, for drug interaction checking, fuzzy medication-name matching, and adherence-history-based reminder optimization [10].

F. Synthesis

The reviewed literature establishes that existing medication adherence systems leave significant gaps in physical confirmation, adaptive AI-driven scheduling, and unified caregiver synchronization, while standards such as ISO/IEC

25010 provide an established framework for evaluating the resulting system holis-tically [11]. GabAIRa Care addresses these gaps by integrating a multi-output XGBoost classifier, REST API-enabled two-way IoT hardware communication, and accessible cross-platform interfaces within a single cohesive system evaluated against that same standard.

Table 1: Summary of Related Literature and Comparison with GabAIRa Care

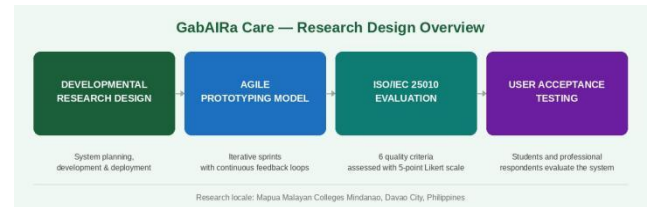
Study / Authors	Year	Approach / Technology	Limitation Addressed by GabAIRa Care
WHO, <i>Adherence to Long-Term Therapies</i> [1]	2003	Global review establishing the ~50% non-adherence baseline among long-term therapy patients	Establishes the problem scale; does not propose a technological intervention
Dalena & Napigkit, Davao hypertensive adherence study [2]	2024	Survey-based study of adherence and lifestyle among hypertensive patients in Davao City	Local-context evidence with no digital or IoT-based remediation
Schnall et al., medication adherence mobile app [4]	2023	Pilot study of a mobile reminder app for medically underserved chronic-illness patients	Notification-only design; lacks physical confirmation and caregiver sync
Chen & Guestrin, XG-Boost [5]	2016	Scalable gradient-boosted tree algorithm for structured data classification	Provides the modeling foundation; not applied to adherence behavior in the original work
Ferreira et al., digital caregiver solutions [6]	2024	Scoping review of digital tools for informal caregivers supporting medication adherence	Identifies fragmentation across tools; no unified patient-caregiver-IoT platform
Alessa et al., AI-based adherence support tools [7]	2025	Focused review of AI-assisted patient support systems for adherence	Reviews AI techniques broadly without a deployed multi-output risk/action model
O'Connor et al., Philippine mHealth adherence support [8]	2022	Mixed-methods study of interactive mobile adherence support in the Philippines	Validates mobile adherence support locally; lacks IoT dispensing integration
Huang et al., universal medication schedule generation [9]	2024	Intelligent scheduling system for generating medication reminder schedules	Addresses scheduling generation only, not adherence risk prediction
Cormen et al., <i>Introduction to Algorithms</i> [10]	2022	Foundational reference for dynamic programming and algorithmic complexity analysis	Provides theoretical basis adopted for GabAIRa Care's scheduling algorithms
ISO/IEC 25010 [11]	2011	Software quality model defining usability, reliability, security, and related criteria	Adopted directly as GabAIRa Care's evaluation framework

was evaluated using ISO/IEC 25010 software quality standards assessing functionality suitability, performance efficiency, usability, reliability, security, and portability.

III. METHODOLOGY

A. Research Design

The study employed a developmental and implementation-oriented research design, following the six phases of the Cross-Industry Standard Process for Data Mining (CRISP-DM) framework to align data science activities with the system's healthcare objectives. **Business Understanding** framed the primary problem as predicting when a patient is at high risk of missing a dose and determining whether a reminder should be sent immediately, delayed, or snoozed, reducing caregiver burden and improving health outcomes. **Data Understanding** involved generating a synthetic dataset of 10,000 patient-session records reflecting realistic adherence behavior across four patient archetypes. **Data Preparation** engineered 12 input features and two target labels, split 80/20 for training and testing using stratified sampling. **Modeling** trained a multi-output XGBoost classifier to simultaneously predict both labels. **Evaluation** assessed the models on the held-out test set using precision, recall, F1-score, and confusion matrices. **Deployment** serialized the trained model and integrated it into the GabAIRa Care backend, exposing a REST API endpoint consumed by the mobile app, the caregiver dashboard, and the ESP32 IoT pill dispenser. The overall system

**Fig. 1:** Research Design Overview

B. Development Methodology

The development of GabAIRa Care followed the Agile Prototyping Model, an iterative approach supporting continuous development, testing, and refinement throughout the project lifecycle. This methodology was selected due to the multi-technology integration requirements of the system, spanning AI, IoT hardware, REST API communication, and cloud synchronization. Development was carried out over five sprints from April to June 2026, beginning with planning and dataset development, progressing through model training, application development, and deployment, with documentation maintained continuously throughout.

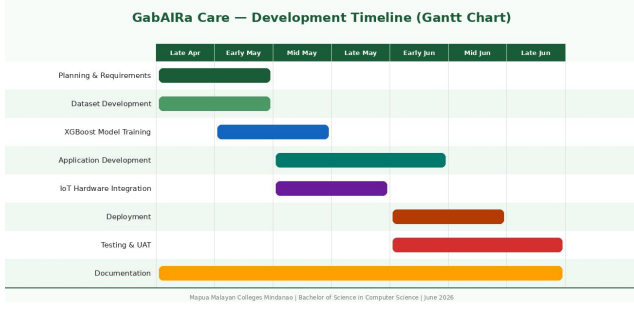


Fig. 2: Development Timeline Across Sprints

C. Dataset Description

The dataset consisted of 10,000 synthetic patient-session records generated using NumPy with a fixed random seed (42) to ensure reproducibility [3]. Four patient archetypes were modeled, consistent (35%), forgetful (30%), irregular (25%), and new (10%), reflecting the distribution of adherence profiles observed in related clinical literature. Twelve features were selected for model training: age, health condition, medication count, notify enabled, suspended, day of week, hour of day, streak_7d, missed last, alert_sent, avg response delay minutes, preferred lead minutes, and cluster label. Two target variables were defined: label risk (0 = Low Risk, 1 = High Risk) and label action (0 = Send Now, 1 = Delay, 2 = Snooze).

All features were numerical or ordinally encoded, requiring no missing-value imputation. Feature engineering steps included computing the 7-day adherence streak from historical logs, deriving the average response delay from timestamped confirmation records, and assigning behavioral cluster labels via k-means clustering on adherence history, yielding four distinct patient profiles consistent with the archetype distribution. The dataset was partitioned into 80% training (8,000 records) and 20% test (2,000 records) sets using stratified sampling with a fixed random state.

D. Model Development

A MultiOutputClassifier from scikit-learn wrapping an XGBClassifier from the XGBoost library was selected because it enables simultaneous multi-label classification from a single model, reducing inference latency suitable for real-time mobile deployment. XGBoost was specifically chosen over alternatives such as Random Forest and Logistic Regression due to its strong performance on structured tabular health data and its support for gradient-boosted trees with regularization, which reduces overfitting on imbalanced action classes [5]. Model hyperparameters were configured as `n_estimators=100`, `max_depth=4`, `learning_rate=0.1`, `eval_metric='mlogloss'`, and a fixed random state. The model was trained on the 8,000-record training set and evaluated on the 2,000-record test set using the standard classification metrics:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

E. Conceptual Framework

Fig. 3 presents the Input-Process-Output framework underlying GabAIRa Care. Patient and medication data, behavioral signal logs, caregiver configuration, IoT pillbox events, and the synthetic training dataset feed into five processing components: the Linear Algorithm, the Dynamic Programming module, the XGBoost Risk and Action Models, and the REST API synchronization layer. These processes generate adherence risk classification, optimized reminder delivery, synchronized dose confirmation, caregiver alerts, and stored adherence analytics.

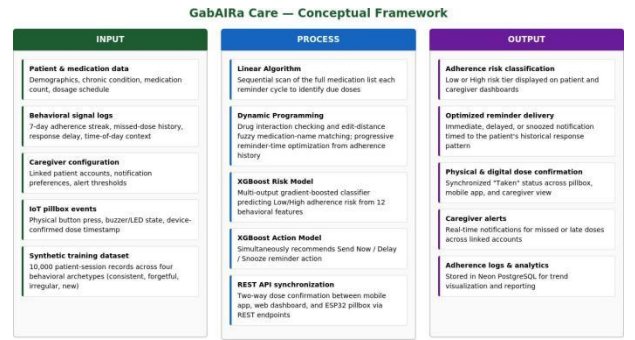


Fig. 3: Conceptual Framework for GabAIRa Care

F. System Architecture

GabAIRa Care is composed of four integrated layers, illustrated in Fig. 4. The **Cross-Platform Application Layer** uses React Native (Expo) for Android, providing the patient interface, caregiver dashboard, schedule view, and adherence statistics. The **AI and Scheduling Layer** hosts the deployed XGBoost model, which serves risk and action predictions via REST API, alongside the Linear Algorithm and Dynamic Programming module. The **Cloud Database Layer** uses Neon PostgreSQL (serverless) to store user profiles, medication schedules, adherence logs, caregiver links, and AI prediction records. The **IoT Hardware Layer** comprises an ESP32-based smart pill dispenser, constructed with a sustainable bamboo enclosure, that dispenses the correct medication compartment and triggers LED and buzzer alerts at scheduled times, synchronized with the application via REST API.

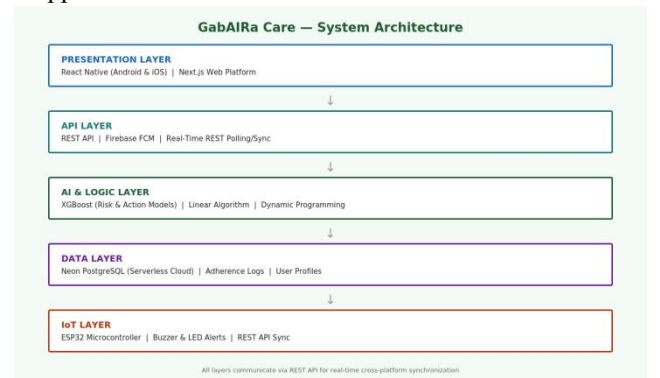


Fig. 4: GabAIRa Care System Architecture

G. IoT Smart Pillbox and Two-Way Synchronization

The ESP32 smart pillbox is equipped with a button, buzzer, OLED display, and LED indicators. When a scheduled dose time is reached, the backend triggers a notification to both the mobile application and the pillbox via REST API, and the pillbox activates its buzzer and LED alert. The patient may confirm intake through one of two methods: pressing the physical confirmation button on the pillbox, or tapping the “Taken” button in the mobile app. Both actions immediately send a confirmation request to the REST API, updating the adherence log in Neon PostgreSQL and notifying the caregiver dashboard in real time, as illustrated in Fig. 6. This two-way design ensures that patients without immediate phone access can still confirm doses physically, while caregivers receive accurate and immediate adherence status across all connected devices. Fig. 5 shows the assembled prototype: a multi-compartment pillbox enclosure housing the wired ESP32 breadboard circuit (left), and a close-up of the OLED status display, physical confirmation button, and illuminated LED alert during a triggered reminder (right).



Fig. 5: GabAIRa Care IoT smart pillbox prototype: full assembly with pill compartments and wired ESP32 circuit (left), and close-up of the OLED display, confirmation button, and active LED alert (right)

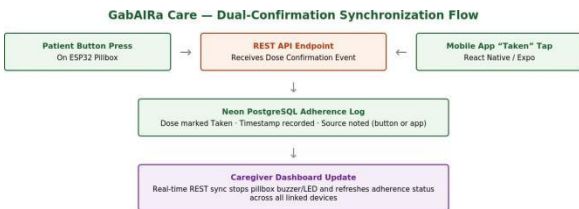


Fig. 6: Dual-Confirmation Synchronization Flow

H. Respondents and Evaluation

Respondents consisted of patients managing chronic illness medications, designated caregivers, and IT experts, recruited via purposive sampling based on direct relevance to the target use case. System evaluation used a structured ISO/IEC 25010-based questionnaire with a five-point Likert scale, assessing

functionality suitability, performance efficiency, usability, reliability, security, and portability across all system components. User Acceptance Testing was additionally conducted with nine (9) respondents, seven students, one nursing student group, and one professional pharmacist, each completing 12 Patient Role test cases (TC-01 to TC-12) and 12 Caretaker Role test cases (IT-01 to IT-12), structured around the use cases shown in Fig. 7.

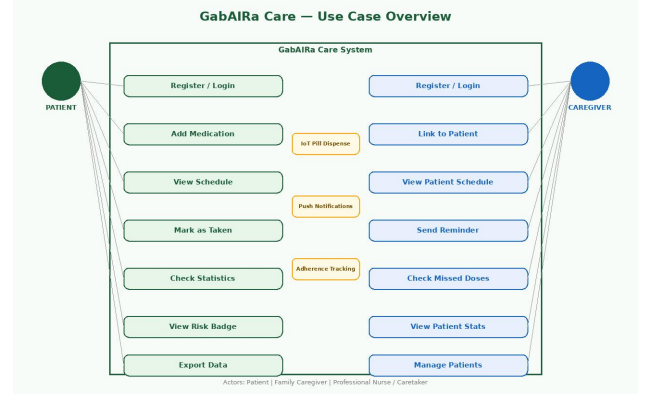


Fig. 7: Use Case Overview for Patient and Caregiver Roles

IV. RESULTS

A. Exploratory Data Analysis

Analysis of the 10,000-record synthetic dataset revealed that the consistent archetype (35%) accounted for the majority of Low Risk labels, while forgetful (30%) and irregular (25%) archetypes contributed disproportionately to High Risk and Delay/Snooze action labels. The missed `last` and `streak_7d` features showed the strongest correlation with `label_risk`, while `avg_response_delay_minutes` and `preferred_lead_minutes` were the most predictive features for `label_action`. Class distribution for the action label reflected realistic imbalance: Send Now (93.6%), Delay (4.6%), Snooze (1.8%).

B. Model Performance

Table II and Table III present the per-class and aggregate performance metrics for the Risk Model and Action Model respectively, evaluated on the 2,000-record held-out test set.

Table II: Risk Model (`label_risk`) – Classification Report

Class	Precision	Recall	F1	Support
Low Risk (0)	1.00	1.00	1.00	1132
High Risk (1)	1.00	1.00	1.00	868
Accuracy		1.00		2000
Macro avg	1.00	1.00	1.00	2000
Weighted avg	1.00	1.00	1.00	2000

Table III: Action Model (label.action) – Classification Report

Class	Precision	Recall	F1	Support
Send Now (0)	1.00	1.00	1.00	1871
Delay (1)	0.98	0.98	0.98	93
Snooze (2)	1.00	0.92	0.96	36
Accuracy		1.00		2000
Macro avg	0.99	0.97	0.98	2000
Weighted avg	1.00	1.00	1.00	2000

The Risk Model achieved perfect classification across both classes with virtually no misclassifications on the test set (confusion matrix: Low Risk, 1131 correct, 1 misclassified; High Risk, 867 correct, 1 misclassified). The Action Model achieved 100% weighted-average accuracy with a macro-average F1 of 0.98. The Snooze class, despite its small support of 36 samples, achieved a recall of 0.92 (33 of 36 correctly classified), reflecting the model’s ability to handle minority classes. Fig. 8 presents the confusion matrices for both models.

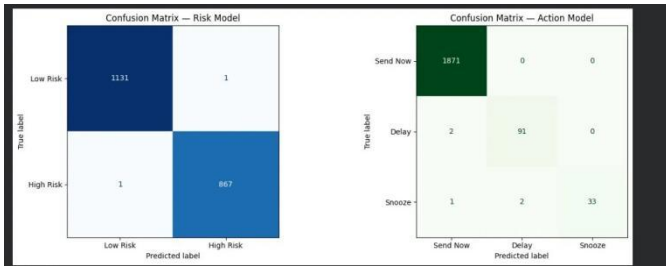


Fig. 8: Confusion matrices for the Risk Model (left) and Action Model (right) evaluated on the 2,000-record test set. The Risk Model shows near-perfect separation between Low and High Risk classes. The Action Model correctly classifies all Send Now instances and achieves high accuracy on the minority Delay and Snooze classes.

C. IoT Synchronization and Algorithm Performance

The real-time two-way synchronization between the GabAIRa Care application and the ESP32 smart pillbox was successfully implemented and demonstrated. Dose confirmation triggered via the physical pillbox button was reflected in the mobile application and caregiver dashboard with minimal latency through REST API communication, and app-side confirmations immediately updated the pillbox status display. The Linear Algorithm consistently identified due medications at each check interval without omissions, and the Dynamic Programming module correctly identified drug interactions across combined medication regimens while successfully reconciling minor medication name spelling variations through fuzzy matching.

D. User Acceptance Testing

User Acceptance Testing was conducted following ISO/IEC 25010 with nine (9) respondents: seven students, one nursing student group, and one professional pharmacist. Each respondent completed 12 Patient Role test cases and 12 Caretaker Role test cases. Table IV summarizes the results.

Table IV: UAT Results Summary

Respondent	Type	Patient	Caretaker
Basil R. Birondo	Student	12/12	12/12
Arl D. Frani	Student	12/12	12/12
Jae Anne Pangan	Pharmacist	12/12	11/12
Adrian J. Rodin	Student	12/12	12/12
Julia N. Alas	Student	12/12	12/12
Jastinne A. Castor	Nsg. Student	12/12	12/12
Katrice B. Moldes	Nsg. Student	12/12	12/12
Jan J. T. Hudes	Student	12/12	12/12
Chlyde C. Espinosa	Student	12/12	12/12
Total		108/108	107/108

All 108 Patient Role test cases passed across all respondents (100% pass rate). The single Caretaker Role failure was IT-07 (Send Reminder to Patient), where pharmacist respondent Pangan observed an approximate 5-minute notification delay. Additional qualitative feedback indicated that the patient-caregiver linking process may be challenging for older users, and that an offline-added patient’s name did not appear on the caregiver dashboard until internet connectivity was restored.

E. ISO/IEC 25010 Software Quality Evaluation

The system received its highest rating in Usability (4.80), reflecting the intuitive interface appreciated by both student and pharmacist respondents. Functionality Suitability (4.56) and Reliability (4.50) were also rated Highly Acceptable. Security received the lowest score (2.69), identifying data protection and HTTPS enforcement as the primary areas for future improvement. Table V presents the full evaluation results.

Table V: ISO/IEC 25010 Evaluation Results

Criterion	Mean	Interpretation
Usability	4.80	Highly Acceptable
Functionality Suitability	4.56	Highly Acceptable
Reliability	4.50	Highly Acceptable
Portability	4.00	Acceptable
Performance Efficiency	3.99	Acceptable
Security	2.69	Moderately Acceptable
Overall Mean	4.09	Acceptable

F. Application Interface and User Acceptance Testing Screens

The following figures illustrate the actual interface of GabAIRa Care as captured during User Acceptance Testing across the Patient, Family/Caregiver, and Caretaker roles.

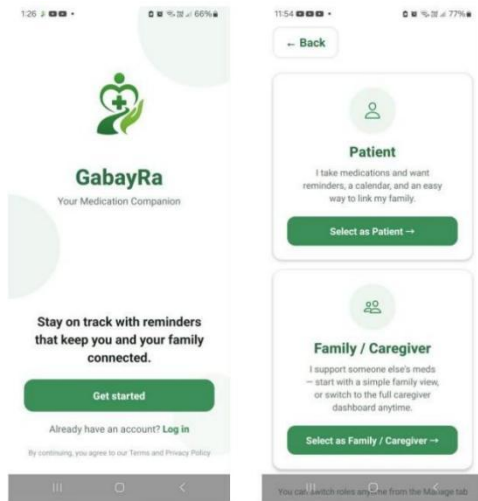


Fig. 9: GabAIRa Care Splash Screen (Left) and Role Selection Screen (Right)

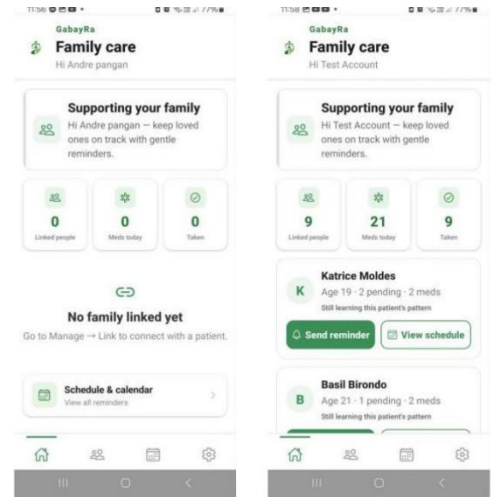


Fig. 12: Family Care Initial Dashboard (Left) and Linked Family View (Right)

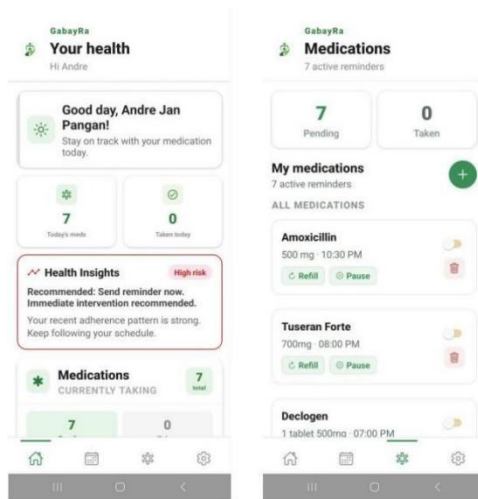


Fig. 10: Patient Home Screen "Your Health" (Left) and Medications List (Right)

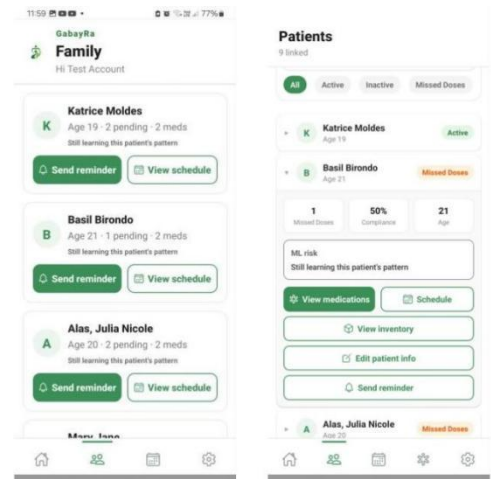


Fig. 13: Family Members List with Reminders (Left) and Patient Management View (Right)

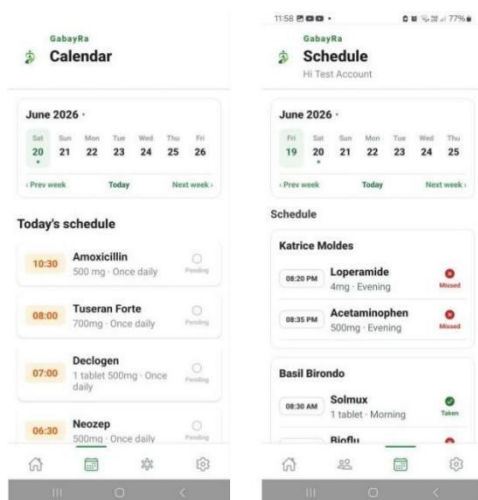


Fig. 11: Patient Calendar View (Left) and Caregiver Schedule View (Right)

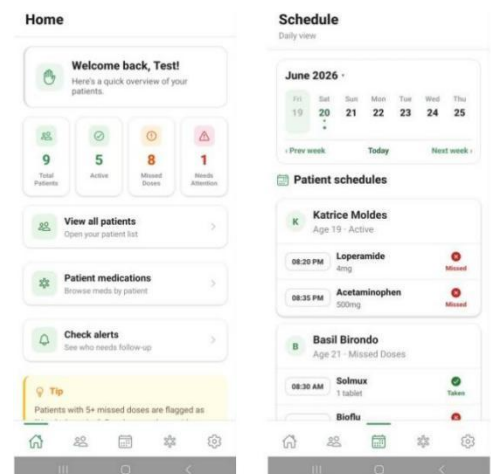


Fig. 14: Caretaker Dashboard with Patient Overview (Left) and Patient Schedules View (Right)

V. DISCUSSION

A. Data–Information–Insight–Action (DIIA) Analysis

Data. The study draws on 10,000 synthetic patient-session records with 12 behavioral and demographic features, two target labels, and adherence logs from the GabAIRa Care mobile application.

Information. The XGBoost model revealed that `missed_last` and `streak_7d` are the strongest predictors of adherence risk, while `avg_response_delay_minutes` and `preferred_lead_minutes` drive the optimal reminder action. The Action Model’s class distribution (Send Now 93.6%, Delay 4.6%, Snooze 1.8%) confirms that the majority of reminders should be dispatched immediately, with a meaningful minority requiring adaptive timing.

Insight. Patients in the forgetful and irregular archetypes benefit most from proactive High Risk classification, allowing the system to escalate notifications before a dose is missed rather than reacting after the fact. The near-perfect Risk Model accuracy (1.00) indicates that behavioral signals reliably distinguish low and high adherence risk within this population.

Action. The system automatically routes High Risk patients to immediate Send Now reminders, dispatches caregiver alerts, and triggers the IoT pill dispenser. Low Risk patients with consistent streaks receive Dynamic Programming-optimized reminders timed to their historically most responsive hours, minimizing notification fatigue while maintaining adherence.

B. Practical Implications

GabAIRa Care demonstrates that a lightweight XGBoost multi-output classifier trained on behavioral features can deliver near-perfect adherence risk prediction suitable for deployment on resource-constrained mobile and IoT devices. The REST API architecture enables seamless synchronization between the mobile app, caregiver dashboard, and ESP32 pill dispenser, making the system practically deployable in home-care settings without requiring dedicated server infrastructure, through Neon Post-greSQL’s serverless model.

The pharmacist respondent’s observation that the caregiver notification system is especially valuable for strictly monitored prescriptions (e.g., anticoagulants, antiretrovirals) underscores GabAIRa Care’s relevance beyond general wellness and into clinical adherence support. The 5-minute notification delay identified in IT-07 points to a known limitation of push-notification pipelines under intermittent connectivity, addressable through more frequent REST polling or persistent real-time channels in future iterations.

C. Limitations and Future Work

The current study has several limitations. First, the XGBoost model was trained exclusively on synthetic data; real-world deployment will require retraining on actual patient adherence records to validate generalizability. Second, the IoT pill dispenser is a prototype and is not certified as a medical device. Third, the system currently requires stable internet connectivity for REST API communication and IoT synchronization; offline-first architecture is a priority for future development. Fourth, the Security criterion scored only 2.69, indicating the need for HTTPS enforcement, end-to-end encryption, and a formal security audit before clinical deployment.

Future work should explore: (1) retraining with longitudinal

real-patient data from Philippine chronic disease cohorts; (2) integrating federated learning to preserve patient privacy while improving model accuracy across sites; (3) expanding IoT hardware to support refill detection, temperature monitoring for medication storage, and multi-compartment intelligent dispensing;

(4) adding Filipino-language UI support to improve accessibility in rural communities; and (5) conducting a controlled longitudinal study measuring actual adherence improvement against a control group.

VI. CONCLUSION AND RECOMMENDATION

This paper presented GabAIRa Care, an XGBoost-assisted cross-platform medication adherence management system that integrates AI-driven risk classification, adaptive reminder scheduling, IoT-assisted manual dispensing, and real-time caregiver synchronization into a unified healthcare solution for patients with chronic diseases.

The multi-output XGBoost classifier simultaneously predicts adherence risk and optimal reminder action with 100% accuracy on the Risk Model and 98% macro-average F1 on the Action Model, evaluated on a 2,000-record held-out test set. User Acceptance Testing demonstrated a 100% Patient Role task completion rate and a 99.1% Caretaker Role pass rate across nine (9) respondents, while ISO/IEC 25010 evaluation yielded an overall mean of 4.09 (Acceptable), with Usability, Functionality Suitability, and Reliability each rated Highly Acceptable.

The system’s modular REST API architecture, cross-platform accessibility, and bamboo-encased IoT smart dispenser position GabAIRa Care as a practical, sustainable, and culturally appropriate medication adherence solution for Filipino patients and caregivers. Future work will focus on real-patient model retraining, offline-first deployment, security hardening, and clinical validation to translate the system’s demonstrated technical performance into measurable health outcomes.

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